Decision Support System for Sri Lankan Paddy Cultivation

AGENDA

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INTRODUCTION

- Sri Lankan paddy agriculture supports national food security and 1.8 million small/medium-scale farmers
- Farmers suffer from low productivity, inefficient water & fertilizer use, disease outbreaks, and income instability due to experience-based decisions
- Cyclone Ditwah (2025) damaged over 100,000 hectares, exposing climate vulnerability
- Existing solutions are fragmented and fail to integrate real-time data, growth stages, disease severity, or market forecasting

✔ **Our Solution:** A unified, cloud-based Decision Support System (DSS) that combines IoT, UAV imagery, ML, and market intelligence for localized, financially efficient paddy cultivation

Research Problem

- How can an integrated Decision Support System using IoT sensors, UAV imagery, machine learning, and market forecasting enable Sri Lankan paddy farmers to make real-time, growth-stage-aware, and economically optimal decisions for irrigation, fertilization, and harvesting?

Research Problem

Objectives

Main Objective

- To design and implement a unified Smart Agriculture Decision Support System that optimizes financial efficiency in Sri Lankan paddy cultivation by combining real-time IoT monitoring, disease prediction, growth-stage resource management, and market-aware recommendations.

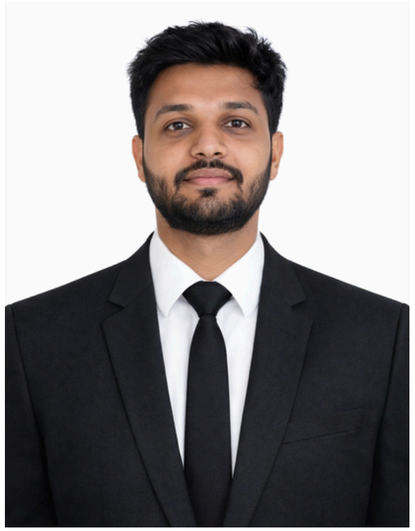
Sub-Objectives

- Develop context-aware IoT field monitoring for real-time soil and environmental interpretation.
- Build intelligent crop recommendation integrating IoT, climate, and market forecasting.
- Create deep learning-based disease severity analysis with loss prediction.
- Implement growth-stage-aware irrigation and fertilization optimization using time-series data and imagery
- Integrate all modules into a farmer-friendly web dashboard with alerts.

Overall System Diagram

IT21811104 | **ASEL SAHANJITH**

Smart Crop Recommendation Using IoT And Market Price Trend



Contribution

This module recommends suitable crops by combining IoT sensor data, climate conditions, and market price trends. It helps farmers choose crops that are both environmentally suitable and economically profitable, and provides recommendations through the central DSS dashboard for better decision-making.

Novelty

Traditional crop recommendation systems rely mainly on static soil and climate data and ignore economic factors. This component introduces a data-driven approach that integrates IoT sensor data, climate conditions, and market prices to recommend crops. The system dynamically analyzes field conditions and market trends to suggest crops that are both environmentally suitable and economically profitable, enabling smarter and more efficient farming decisions.

Problem Definition

Sri Lankan farmers often select crops using traditional guidelines, ignoring real-time soil, climate, and market conditions, leading to lower yields, wasted resources, and reduced income. A data-driven crop recommendation system is needed to optimize environmental suitability and profitability.

Research Objective & Sub-Objectives

Main Objective

To **design and validate an IoT- and AI-based crop recommendation system** that collects real-time soil and climate data, analyzes market price trends, **trains adaptive models**, and integrates with a user-friendly dashboard to provide environment- and profit-optimized crop selection guidance for farmers.

Sub-Objectives

- IoT-Based Data Collection
- Market Data Integration
- Machine Learning-Based Recommendation Engine
- User Interface / Dashboard

Research Methodology

Knowledge domain that contributes to methodology

- Data Collection → Collect IoT soil sensors, Climate data ,Market prices
- Data Preprocessing → Extract features and generate suitability and price trend indicators
- Modeling → ML for crop suitability and integrate economic viability
- Recommendation → specific crop suggestions and dynamic updates real-time data
- Dashboard → Farmer-friendly and visualization, soil, climate, recommended crops
- Validation → Test real farms and evaluate accuracy

Specific Latest Technologies utilized

Market Data APIs & Predictive Analytics with trend analysis, and profit prediction
IoT Sensors nutrient and temperature sensors for real-time field monitoring
Machine Learning & AI Models Neural Networks, and Decision Trees
Data Analytics & Visualization Tools like Power BI, Tableau, or web dashboards

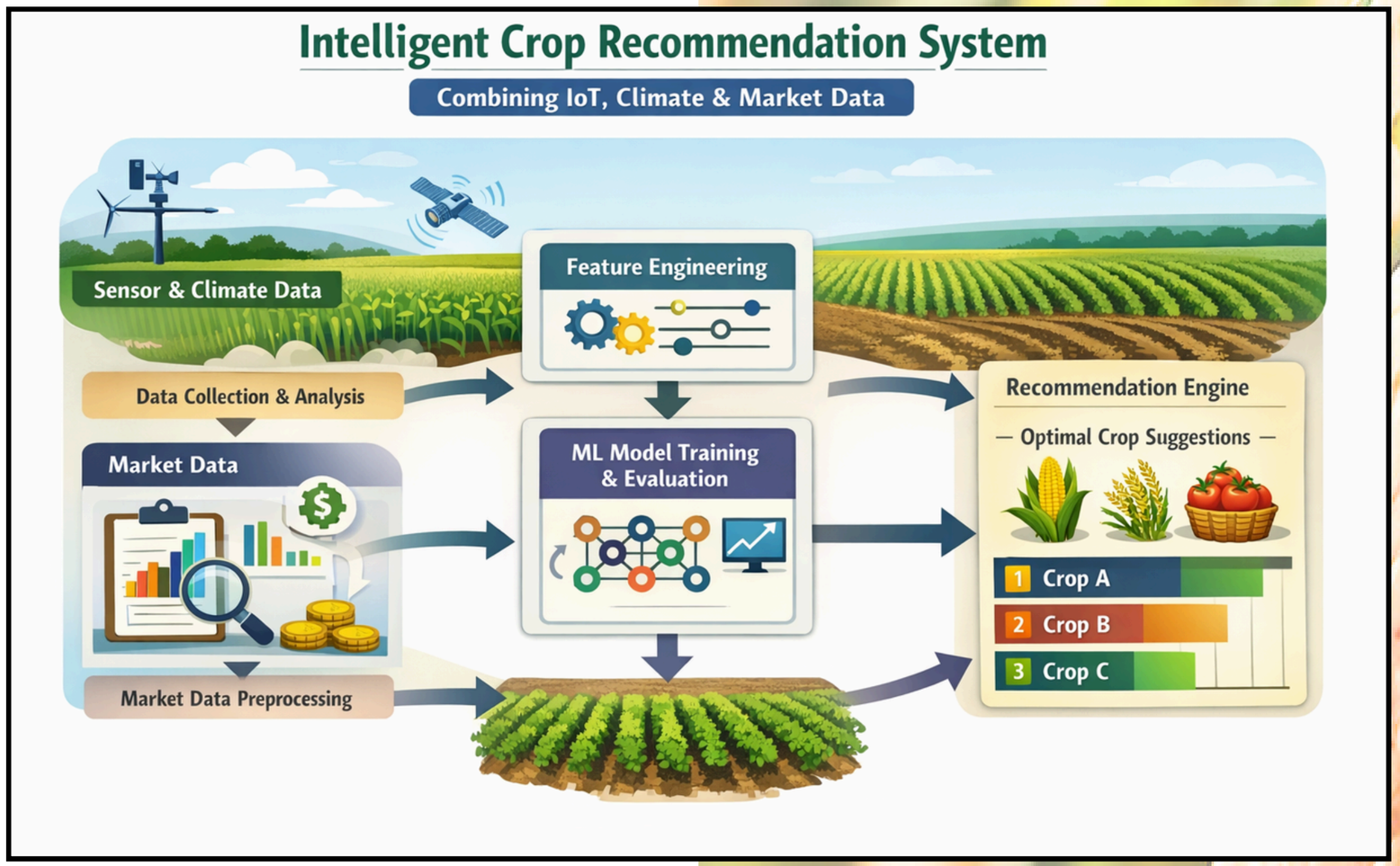
How to Validate and Evaluate

- Recommendation Accuracy : Use past data to check whether the crops recommended by the system would have performed well.
- Crop suitability accuracy: Check Percentage of recommended crops suitable for the soil and climate.

Data Availability, Security & Ethical Clearance

Data Availability → Real-time IoT sensors monitoring soil and field conditions + UAV images of paddy fields
Validate → On-device processing of sensor data ,HTTPS for secure cloud communication
Ethical → Data anonymization to protect privacy

System Overview of the Individual Component



Smart Crop Recommendation Details

Dependencies and Common Components

- Depends on IoT Field Monitoring Module for soil and environmental data.
- Depends on Market Price Analysis Module for profitability evaluation.
- Depends on external agricultural market data APIs for real-time crop prices.

Common Component →
Central cloud database + Data communication API + Unified farmer dashboard (web application)

IoT Device Integrated Field Monitoring System

Contribution

This growth-stage adaptive sensor framework continuously monitors soil and environmental conditions, then delivers actionable, variety-specific recommendations directly to farmers' individual dashboard when it comes to selecting best variety, optimal fertilization and optimal disease remedies which directly enhances the yields.

Novelty

- We are introducing first growth-stage adaptive sensor devices that recommend optimal paddy variety, best fertilizer type and times with reminders and disease stage awareness alerts showcasing proof of integrating every aspect from irrigation to yield.

Problem Definition

- Sri Lankan paddy farmers lack real-time data support, leading to wrong paddy choices, poor fertilizer choices and disease prevention failures leading to 10-20% growth yield loss.

Research Objective & Sub-Objectives

Main Objective

- Develop a context-aware module that optimizes irrigation & fertilization recommendations using time-series sensor data and imagery.

Sub-Objectives

- Analyze real-time soil sensor data for nutrient and moisture patterns
- Predict fertilization recommendation using soil data patterns.
- Create reference data to analyze disease stages.
- PWA Dashboard recommending actions.

Research Methodology

Knowledge domain that contributes to methodology

- Design Science Research + IoT Systems + Data Analytics (SST – Software Systems & Technologies)
- Knowledge domain: Agronomy + Relational Databases + IoT for precision agriculture

Specific Latest Technologies utilized

- ESP32 microcontrollers with multi-parameter sensors for real-time field data gathering.
- Python with FastAPI for data preprocessing pipeline and noise filtering
- React-based Progressive Web App for dashboard.
- RESTful APIs for seamless integration with disease detection and market modules

Research Methodology

How to Validate and Evaluate

- Field validation across 6 farmer fields in multiple climatic zones for one full **Yala Season.**
- Compare recommendation accuracy with Department of Agriculture guidlines.
- Assess usability feedback from farmers

Data Availability, Security & Ethical Clearance

- Data: Real-time IoT sensors + Paddy varieties(RRDI-Bathalagoda)
- Security: Device Physical Security + Protected Backend using VPS KVM01.
- Ethical: Farmer consent forms + data anonymization + SLIIT Ethics Committee clearance

Complietion of the Project

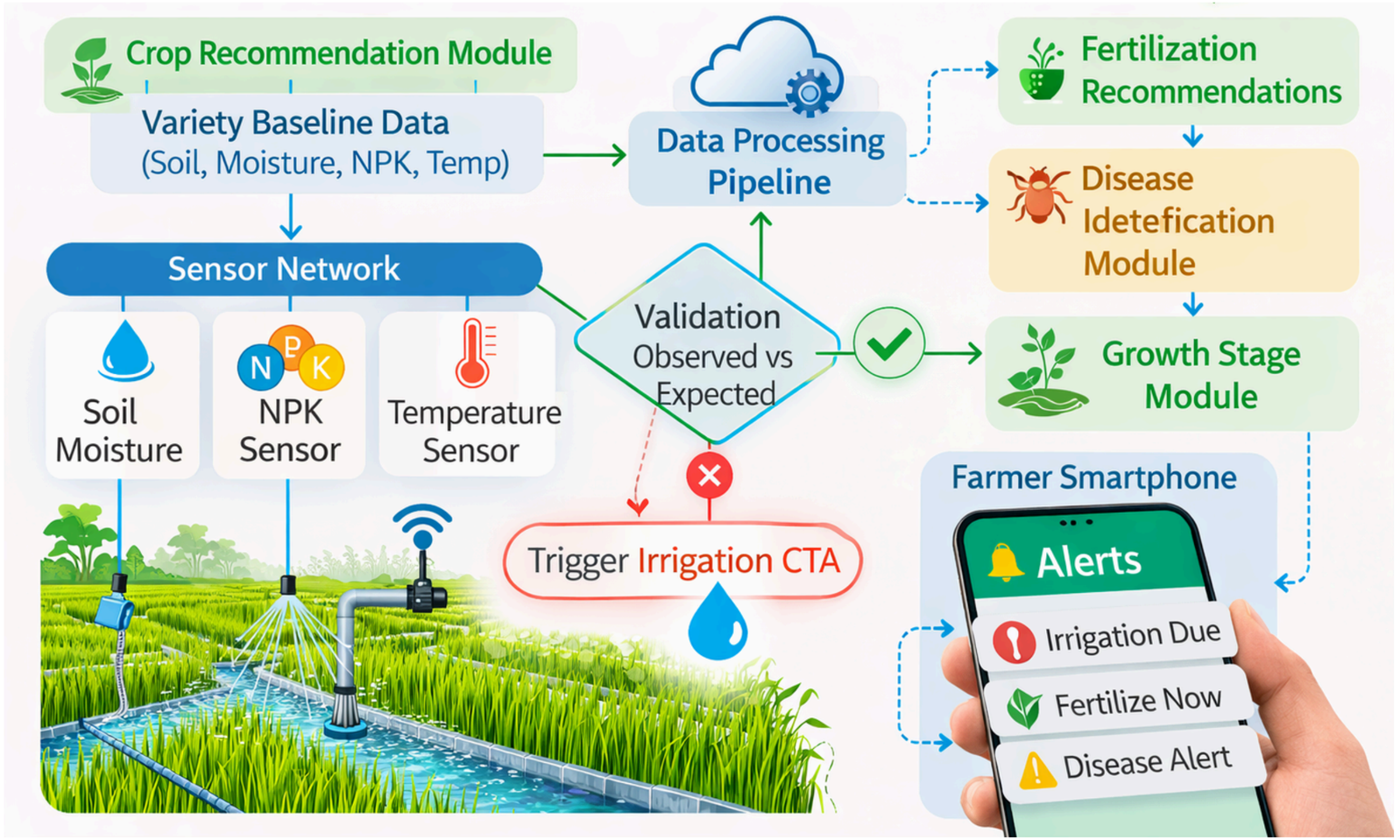
System Overview of the Individual Component

- Obtains real-time soil data from IoT sensor network
- Preprocess data and compare with Crop recommendation data.
- Interprets data for growth stage fertilization & disease recommendation.
- Dashboard alerts & recommendations

Dependencies and Common Components

- Depends on Market Recommendation module.
- Shares output with Disease Analysis & Fertilization Recommendation modules
- Common components: Central cloud database + Unified farmer dashboard (web app)

System Overview of the Component



Growth Stage-Aware Irrigation and Fertilization Optimization

Contribution

This module predicts paddy growth stages using time-series sensors and UAV imagery, then generates dynamic irrigation and fertilization schedules. It reduces resource waste by 20–30% and feeds real-time recommendations into the central DSS dashboard.

Novelty

- Traditional irrigation & fertilization use fixed generalized schedules or static rules. This module introduces a predictive, growth-stage-aware optimization mechanism using time-series sensor data and crop development indicators. Resource recommendations dynamically adapt to plant stage, soil conditions & environmental variability – replacing fixed-rule farming with precision, stage-sensitive decision-making.

Problem Definition

- Sri Lankan paddy farmers waste 20-40% of water & fertilizer because they follow generalized schedules without considering real-time growth stages.

Main Objective

- Develop a growth-stage-aware module that optimizes irrigation & fertilization resources using time-series sensor data and imagery.

Sub-Objectives

- Analyse time-series data
- Predict stages from images/temporal data
- Model stage-specific needs
- Generate dynamic schedules
- Dashboard integration

Research Methodology

Knowledge domain that contributes to methodology

- Design Science Research + Time-Series Analysis + Computer Vision (SST – Software Systems & Technologies)
- Knowledge domain: Agronomy + Machine Learning + IoT for precision agriculture

Specific Latest Technologies utilized

- Time-series forecasting with LSTM + Attention mechanism (PyTorch)
- Crop stage prediction using CNN-LSTM on UAV/multispectral imagery
- Leaf Area Index (LAI) estimation with CNN-LSTM-Attention model
- Dynamic scheduling using Gaussian Process Regression for phenology

How to Validate and Evaluate

- Metrics: RMSE, Accuracy (>95%), Water & Fertilizer saving % (target 20-30%)
- Field trials in Sri Lankan paddy farms vs traditional method
- Comparison with static schedules (real yield & cost data)

Data Availability, Security & Ethical Clearance

- Data: Real-time IoT sensors + UAV images from local farms (Department of Agriculture collaboration)
- Security: On-device processing + AES encryption + cloud HTTPS
- Ethical: Farmer consent forms + data anonymization + SLIIT Ethics Committee clearance

Completion of the Project

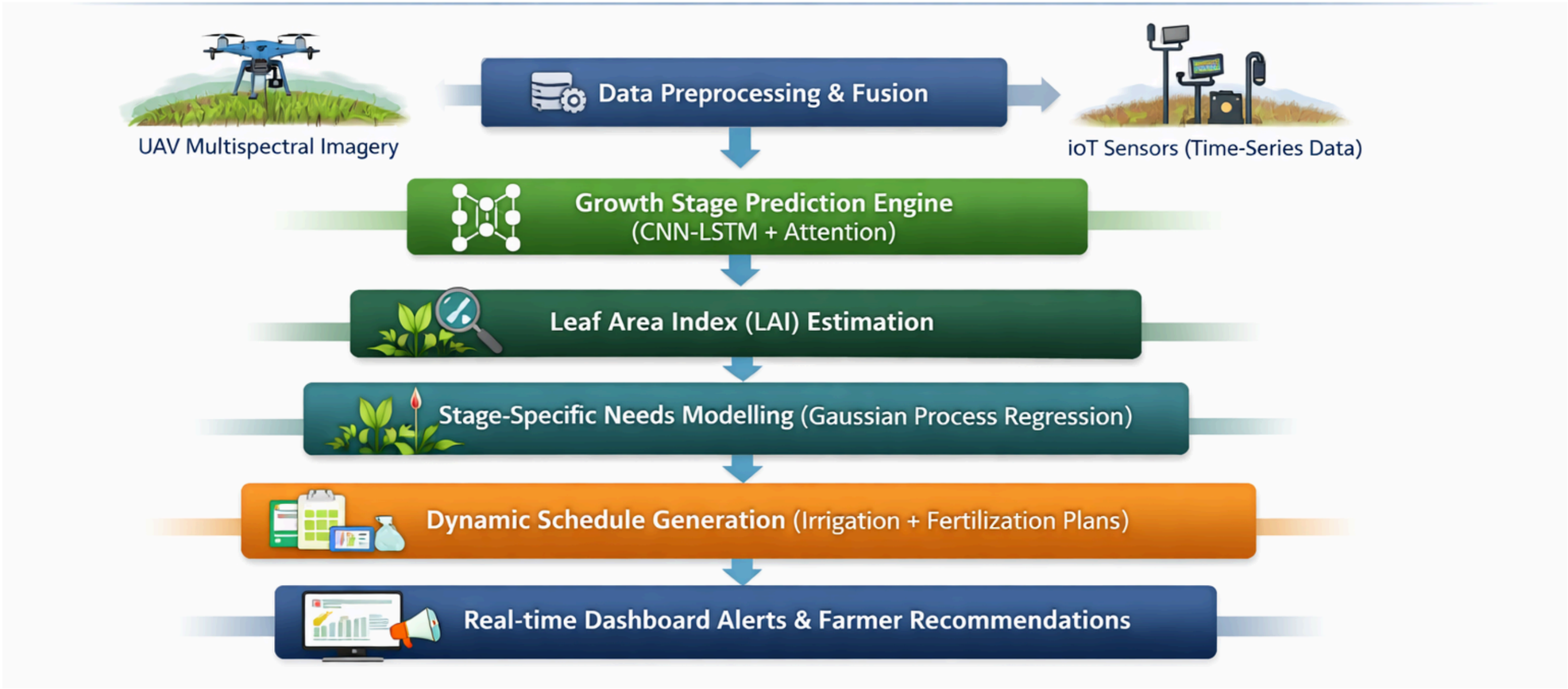
System Overview of the Individual Component

- Time-series sensor data (soil moisture, NPK, temperature, humidity) + UAV imagery
- Stage Prediction Engine (CNN-LSTM)
- Needs Modelling (stage-specific water & fertilizer calculator)
- Dynamic Schedule Generator
- Dashboard alerts & recommendations

Dependencies and Common Components

- Depends on IoT Field Monitoring module (real-time data feed)
- Shares output with Disease Analysis & Market Recommendation modules
- Common components: Central cloud database + Unified farmer dashboard (web app)

System Overview of the Individual Component



Disease Detecting and Serverty Estimation System

Contribution

This module predicts disease severity stages using paddy leaf images and environmental time-series, then generates dynamic treatment schedules with quantified harvest loss estimates. It prevents yield losses exceeding 30% by delivering timely, risk-informed recommendations and feeds real time serverity alerts.

Novelty

- This component advances the state of disease severity estimation and progression awareness using both image analysis and environmental factors. The system moves beyond “disease present/absent” outputs to provide risk-informed management recommendations . Furthermore, localization for Sri Lankan paddy diseases and climatic conditions enhances practical reliability compared to generic datasets and image only models.

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Problem Definition

- Sri Lankan paddy farmers face critical economic threats from disease outbreaks such as rice blast (Magnaporthe oryzae) and brown spot (Bipolaris oryzae), which cause yield losses exceeding 30% in severely affected seasons.

Research Objective & Sub-Objectives

Main Objective

- Develop an integrated diseases serverity estimation and harvest loss prediction module.

Sub-Objectives

- Collect and label Sri Lankan paddy leaf images
- Train deep learning models for severity classification.
- Integrate real-time humidity, temperature and leaf wetness data.
- Generate disease progression forecast and harvest loss estimates.
- Deliver actionable management recommendations.

Research Methodology

Knowledge domain that contributes to methodology

- Knowledge domain that contributes to methodology
- Design Science Research + Computer Vision + Multimodel Deep Learning + Time Series Analysis (SST – Software Systems & Technologies)
- Knowledge domain: Agronomy + Machine Learning + IoT Sensor Fusion for precision paddy disease Management.

Specific Latest Technologies utilized

- Disease severity estimation and progression forecasting using EfficientNet-B4 + LSTM-Attention multimodal model (PyTorch).
- Multimodal fusion of leaf images and real-time IoT data (humidity, wetness, temperature) using CNN + MLP with cross-attention mechanism.
- Harvest loss timeline prediction using CNN-LSTM-Attention regression head on sequential severity and environmental time-series.
- Actionable recommendation generation using ensemble learning (severity thresholds + rule-based inference) integrated with Department of Agriculture guidelines.

How to Validate and Evaluate

- MAE < 0.5 grades — T+7 / T+14 progression forecast vs. observed field outcomes
- Harvest-Loss Correlation $r^2 > 0.80$ — estimated vs. actual yield reduction in trial plots
- System Latency ⩽ 3 sec — image upload to full recommendation output on dashboard.

Data Availability, Security & Ethical Clearance

- Data: Paddy leaf images, public datasets (PlantVillage & IRRI), real-time IoT sensor feeds, and HARTI yield-loss records support model training and validation.
- Security: Farmer data is protected through HTTPS, AES-encrypted cloud storage.
- Ethical: Farmer consent forms, full data anonymization, and SLIIT Ethics Committee clearance.

Completion of the Project

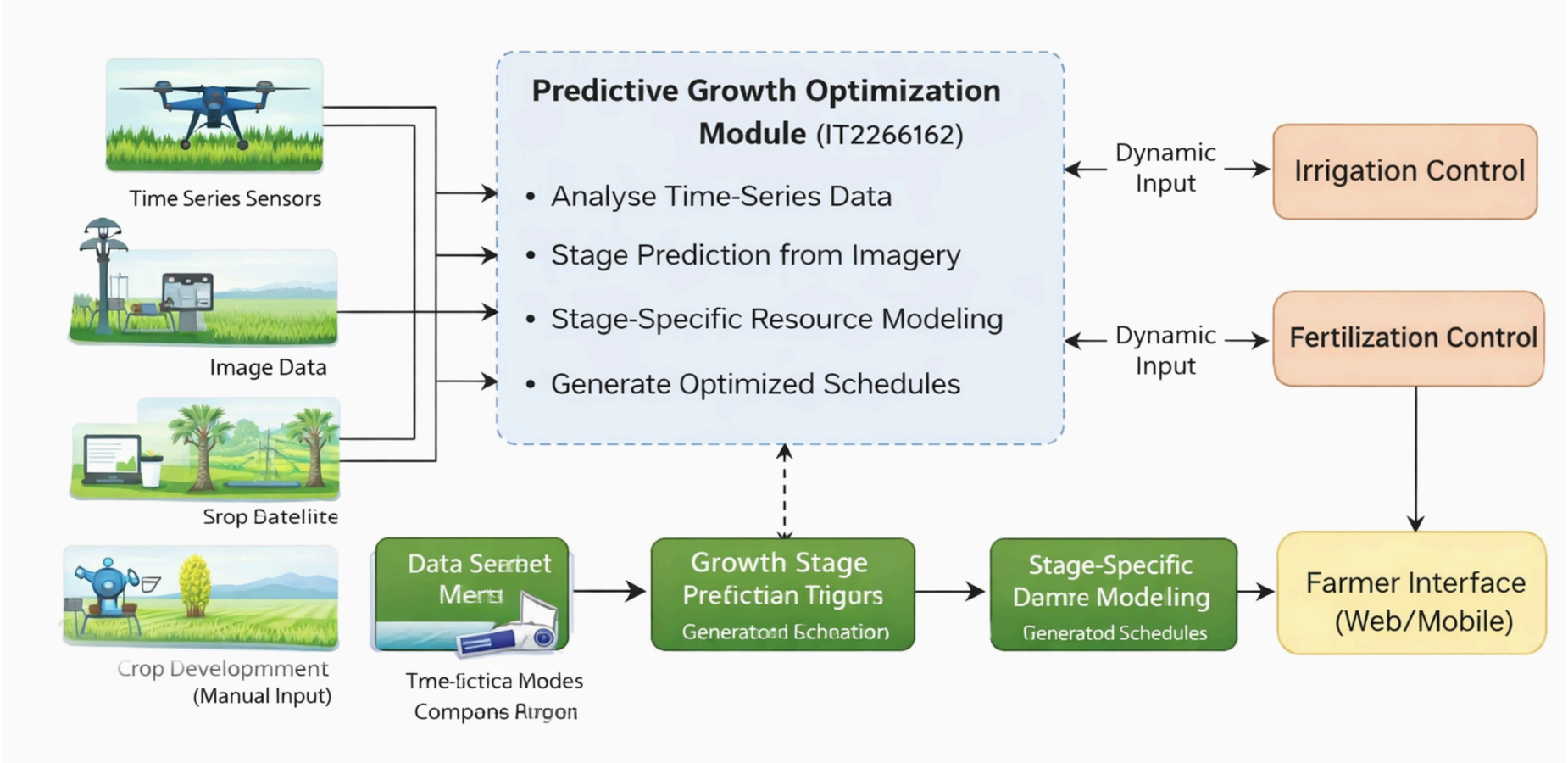
System Overview of the Individual Component

- Time-series sensor data (soil moisture, temperature, humidity) + CAM imagery.
- Disease Classification Engine (EfficientNet-B4 + CNN with Grad-CAM).
- Severity & Progression Modelling.
- Harvest-Loss Estimation Engine(severity grade × crop growth stage × time-to-treatment → yield-loss percentage).
- Dashboard alerts & recommendations.

Dependencies and Common Components

- Depends on IoT Field Monitoring module for real-time data.
- Shares disease output & harvest-loss estimates with Growth Stage Optimization module.
- Common components: Central cloud database + Unified farmer dashboard.

System Overview of the Individual Component



References

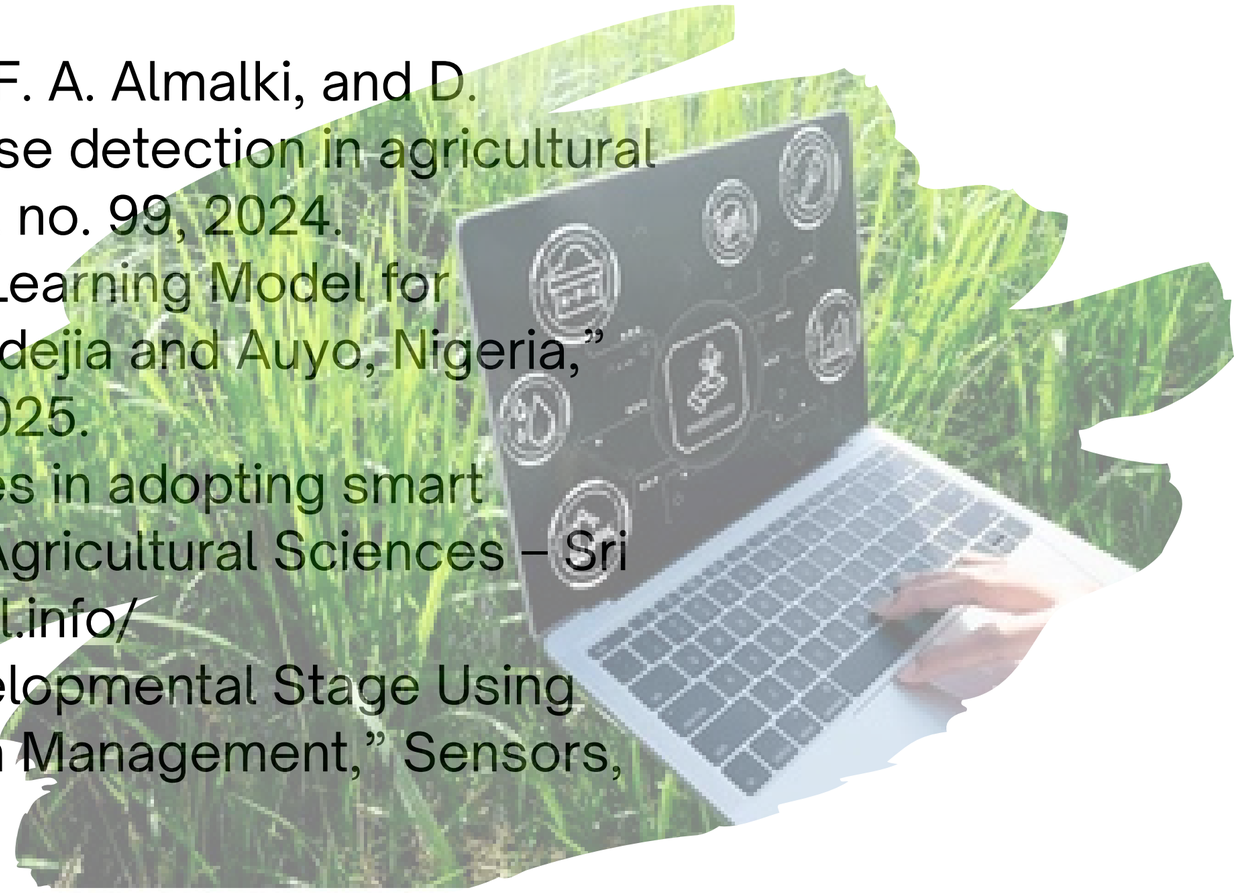
[1] Department of Agriculture – Sri Lanka, “Crop Recommendations and Fertilizer Guidelines,” 2021.

[2] Y. Wang, U. S. R. Dhamodharan, N. Sarwar, F. A. Almalki, and D. Mohan, “A hybrid approach for rice crop disease detection in agricultural IoT system,” Discover Sustainability, vol. 5, art. no. 99, 2024.

[3] I. Abdurrahman and A. M. Miyim, “Machine Learning Model for Predicting Rice Crop Yield: A Case Study in Hadejia and Auyo, Nigeria,” UMYU Scientifica, vol. 4, no. 1, pp. 239–249, 2025.

[4] D. Pathirana and W. Wijesinghe, “Challenges in adopting smart farming technologies in Sri Lanka,” Journal of Agricultural Sciences – Sri Lanka, 2020. [Online]. Available: https://jas.sljol.info/

[5] C. Y. Yang et al., “Assessment of Rice Developmental Stage Using Time Series UAV Imagery for Variable Irrigation Management,” Sensors, vol. 20, no. 18, p. 5354, 2020.



Thank You